Source Code and Data Description

Evaluating and Reducing Hallucinations in LLM-Based Medical Report Summarization

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1. Repository at a glance

Item	Value
Source	https://github.com/slay-a/medical-hallucination-eval (public, branch main)
Commit	2906a52 (2026-11-06)
Type	Python 3 research codebase; Bash helper scripts; no compiled components
Size	code 919 KB; result artifacts 6.4 MB
Language and runtime	Python 3.9.6 in a virtual environment (.venv) for the pipeline; Python 3.13 in .venv-llm for the mlx-lm model server; macOS on Apple silicon (M5, 24 GB). The pilot runs on the CPU; the judge study, MedNLI fine-tuning and main study use the GPU through PyTorch Metal and MLX
License / use	Academic use. MTSamples-derived results are public. Outputs derived from PhysioNet data are never committed (results_private/ is git-ignored).
Reproducibility	Seeds: 42 (MTSamples sample, MedNLI fine-tuning, repeated-generation subset, selection-bias analyses), 1000r + k (repeated generations), 7 (audit sample) and 8 (its wave-2 extension), 0 (pilot bootstrap); pinned requirements and lock files; every pilot number is reproducible offline from results/claims_all.csv with recompute_metrics.py; main-study numbers require the credentialed data and are reproduced by run.sh --main-study

2. File inventory

Path	Size	Purpose
hallucination_eval.py	27 KB	Main pipeline: loads MTSamples, samples 50 notes (seed 42), generates E0 (full note) and E1 (top-3 retrieved chunks only) summaries with GPT-4o-mini, segments claims (spaCy; header lines removed), retrieves top-3 evidence sentences (all-MiniLM-L6-v2), labels claims with cross-encoder/nli-MiniLM2-L6-H768, writes per-claim and per-summary results.
e2_extractive_eval.py	20 KB	E2 extractive baseline: centroid ranking of source sentences by mean cosine similarity, top-5 kept in document order, no LLM call; evaluated with the same judge; appends E2 rows and columns to the result files.
e1b_fullnote_rag_eval.py	4 KB	E1b: RAG with the full note plus the retrieved excerpts, and its evaluation. Requires OPENAI_API_KEY; 50 calls.
e3_cove_eval.py	10 KB	E3: factored Chain-of-Verification on top of the E1 draft; per-claim verification against top-3 sentences from the full note, then revision (keep / replace / remove per claim); checkpointed with a --revise-only

Path	Size	Purpose
		mode. Requires OPENAI_API_KEY; about 500 calls.
preprocessing.py	4 KB	Shared utilities: is_markdown_header() removes section-header lines from the claim set; clean_mtsamples_text() and sentencize_cleaned() repair the comma-encoded line breaks of MTSamples.
recompute_metrics.py	11 KB	Offline recomputation of every metric from results/claims_all.csv with headers excluded: per-sample UFR/CR, comparison table, aggregate statistics, Wilcoxon tests, bootstrap 95% CIs, effect sizes, header-filter effect tables.
ablations.py	15 KB	Offline robustness analyses: NLI threshold sweep, evidence top-k (1, 3, 5), cleaned-evidence re-scoring, coverage proxy, negation analysis, keyword-assisted error taxonomy.
calibration_annptsumm.py	12 KB	Judge validation against ann-pt-summ expert annotations (credentialed data, local models only): sentence-level precision/recall/F1/kappa, AUROC, threshold sweep, per-label recall, summary-level correlation.
calibration_variants.py	7 KB	Five evidence-aggregation variants of the judge evaluated on the expert-annotated set.
judge_candidates.py	14 KB	Judge selection study: six candidate judges (MiniLM, two DeBERTa-v3-large NLI models, MiniCheck DeBERTa and RoBERTa, Bespoke-MiniCheck-7B) plus the MedNLI-tuned model, each in two evidence modes (top-3 sentences; whole course in 400-token windows), scored against the 1,781 expert-annotated sentences with thresholds chosen on the other subset; writes results/judge_candidates.csv.
minicheck_server.py	7 KB	Local HTTP scoring server for Bespoke-MiniCheck-7B (InternLM2.5-7B grounding checker, 8-bit MLX conversion kept outside the repository); returns the probability of 'Yes' for (document, claim) pairs with per-document prompt caching; queried by judge_candidates.py and mimic_evaluate.py over localhost.
finetune_mednli.py	8 KB	Fine-tunes the strongest general NLI model on MedNLI (1 epoch, Adafactor, lr 2e-5, effective batch 16 as 4 x 4 accumulation, max length 128, embedding block frozen, MPS); saves the best-dev checkpoint outside the repository (~/Desktop/Thesis/models).
mimic_generate.py	14 KB	Main-study generation against the local OpenAI-compatible mlx-lm server (Qwen2.5-7B-Instruct, 4-bit): E0, E1 (with citations) and its six retrieval variants, E1b, E2 and E3 for the 110 ann-pt-summ hospital courses; resumable JSONL output in results_private/. With the MIMIC_TAG environment variable, the same script runs the generator-robustness reruns (Meta-Llama-3.1-8B-Instruct-4bit, tag llama31; Mistral-7B-

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		Instruct-v0.3-4bit, tag mistral7b) into tagged files; for a chat template that rejects a separate system role, the system prompt is folded into the user message.
mimic_extra.py	6 KB	Main-study variants that separate confounds: E0 with a length instruction of about 150 words, E1b without the citation instruction, and two repeated generations of E0, E1, E1b and E3 on 40 courses (seeds 1000r + k); appended to the private generation file.
citations.py	3 KB	Shared claim splitter: removes citation markers before spaCy segmentation, assigns each marker to its sentence by character position, and strips section headers merged into a sentence.
mimic_evaluate.py	17 KB	Main-study evaluation with the judge of highest kappa: claim splitting (citations.py), forward UFR/CR, supported and unsupported counts, citation coverage, accuracy and reliability, reverse coverage of the clinician's discharge instructions and the number of their sentences not covered, scoring of the doctor-written instructions themselves, paired Wilcoxon tests with Holm adjustment over the primary family, ablation tests; aggregates to results/mimic_*.csv, per-claim rows to results_private/.
atomic_claims.py	12 KB	Atomic-fact analysis: the local model splits every expert-set sentence and every main-study claim into atomic facts; each fact is scored by the selected judge; agreement with the experts and main-study rates at the level of facts (results/atomic_results.csv).
judge_selection_bias.py	7 KB	Optimism and independence of the judge evaluation: nested cross-validation of the whole selection procedure by course, patient-disjoint transfer between the doctor-written and generated summaries, course-clustered bootstrap intervals and per-source agreement.
data_overlap.py	5 KB	Matches every hospital course to its MIMIC-IV-Note discharge note by normalized text snippets and counts patient and admission overlap between the main study and the expert set (counts only).
mimic_revision_analysis.py	16 KB	Variant tests, run-to-run variation, abstention analysis, coverage equivalence check, question-answering utility, per-condition bootstrap intervals, comparison of the first and the corrected claim splitter, and the overlapping-window check.
check_no_protected_text.py	8 KB	Scans every tracked file, including PDF, Word and PowerPoint text, for eight-word sequences of the credentialed texts and of the private outputs; writes counts only.
mimic_second_judge.py	6 KB	Second-judge agreement analysis: scores every base-condition claim with Bespoke-MiniCheck-7B (best mode and threshold from the judge study) and reports per-

Path	Size	Purpose
		condition agreement, kappa, confirmed flags and UFR under both or either judge (results/mimic_judge_agreement.csv).
audit_sample.py, audit_tool.py, audit_summary.py, audit_agreement.py	11 KB + 15 KB + 9 KB + 3 KB	Blind manual audit of the judge in two passes: course-clustered stratified sample and its wave-2 extension (60 courses, 340 claims, 60 citation checks; seeds 7 and 8), a local labeling page on 127.0.0.1 that hides the condition and the judge's label, weighted estimates of the judge's precision, false omission rate, sensitivity, specificity and kappa and of the manually confirmed unsupported share, and the agreement between the two labeling passes (results/audit_*.csv).
verifact_eval.py	23 KB	External check of the selected judge on MIMIC-III-Ext-VeriFact-BHC (PhysioNet, credentialed): scores the clinician-adjudicated propositions against the five best-matching 400-token windows of the summarized admission's notes (the admission's notes run to hundreds of windows) with the main study's windowing and threshold, checks the cap against all-window scoring on the shortest admissions, and computes agreement with the clinician labels and the published VeriFact raters' baseline (results/verifact_*.csv; runs locally, resumable).
mimic_qa.py	7 KB	Question-answering pilot: three fixed questions per course answered from the full course (QA-E0) or three retrieved chunks (QA-E1) with an abstention instruction; evaluation of abstention rate, UFR/CR and support by the clinician's instructions.
analyze.py	27 KB	Figures (results/fig_*.png, 200 dpi) and example tables (claims fixed by RAG, persistent unsupported claims).
thesis/build_thesis.py	4 KB	Builds the thesis (.docx and .pdf) in CSUN format from the result files; content modules ch1_4.py, ch5_pilot.py, ch_judge.py, ch_mimic.py, ch8_9.py, appendices.py; two-pass table of contents, list of tables and list of figures via LibreOffice.
thesis/render_docx.py, thesis/results_loader.py, thesis/references.py	27 KB + 8 KB + 21 KB	Rendering engine (python-docx), results access layer, IEEE reference database with citation resolver.
deliverables/build_progress_slides.py	33 KB	Progress report slides (PDF and PPTX).
deliverables/ build_source_data_description.py	30 KB	This document.
setup.sh, run.sh, requirements.txt, requirements-llm.txt	2 KB + 4 KB + 0 KB + 0 KB	Environment bootstrap (setup.sh --llm also creates the model-server environment); reproduction chains (bash run.sh, bash run.sh --offline, bash run.sh --main-study); pinned dependencies, with complete lock files requirements-lock.txt and requirements-llm-lock.txt.

3. Dependencies

requirements.txt (verbatim): # Pipeline environment (.venv, Python 3.9). Exact versions used for every reported result;; # requirements-lock.txt lists the complete environment including transitive dependencies.; openai==2.31.0; pandas==2.3.3; numpy==1.26.4; spacy==3.7.4; sentence-transformers==5.1.2; transformers==4.57.6; torch==2.8.0; scikit-learn==1.6.1; scipy==1.13.1; matplotlib==3.9.4; tqdm==4.67.3; python-docx==1.2.0; reportlab==4.4.10; python-pptx==1.0.2; rank-bm25==0.2.2; sentencepiece==0.2.2; protobuf==6.33.6; pillow==11.3.0.

Installed versions used for the reported results: openai 2.31.0, pandas 2.3.3, numpy 1.26.4, spacy 3.7.4 (en_core_web_sm), sentence-transformers 5.1.2, torch 2.8.0, scikit-learn 1.6.1, scipy 1.13.1, matplotlib 3.9.4, python-docx 1.2.0, reportlab 4.4.10, python-pptx 1.0.2. LibreOffice (soffice) converts .docx to PDF.

External models (downloaded once from the Hugging Face hub, then cached): sentence-transformers/all-MiniLM-L6-v2 (bi-encoder, 384-d, retrieval by cosine similarity); cross-encoder/nli-MiniLM2-L6-H768 (NLI classifier; outputs contradiction/entailment/neutral). External API: OpenAI GPT-4o-mini, temperature 0.3, max 450 output tokens; key supplied via the OPENAI_API_KEY environment variable and never stored in the repository. Judge study and main study: MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli, cross-encoder/nli-deberta-v3-large, lytang/MiniCheck-DeBERTa-v3-Large and lytang/MiniCheck-RoBERTa-Large (candidate judges); mlx-community/Qwen2.5-7B-Instruct-4bit served with mlx-lm 0.31 (mlx 0.32) through an OpenAI-compatible interface on localhost; bespokelabs/Bespoke-MiniCheck-7B converted to 8-bit MLX and served by minicheck_server.py (judge candidate); rank-bm25 for the BM25 ablation. No protected text is sent to any external service.

4. Data sources

Source	Type	Content	Size	Access status and location
MTSamples	Public CSV	4,999 sample medical transcriptions, 40 specialties; 624 eligible Discharge Summary and Consult - History and Physical notes; 50 sampled (seed 42): 40 consultations, 10 discharge summaries; source length 117 to 1,809 words (mean 539); summaries generated under five conditions	17 MB	Public. Stored one directory above the repository (../mtsamples.csv), not tracked.
MIMIC-IV-Note v2.2	Credentialed CSV (gzip)	331,794 de-identified discharge summaries and about 2.3 million radiology reports (PhysioNet doi 10.13026/1n74-ne17)	1.8 GB compressed	Credentialing approved 4/28/2026; downloaded 5/14/2026; gzip integrity verified; SHA-256 manifest present. Stored locally (~/Desktop/Thesis). Used only to match each hospital course to its discharge note for the patient-overlap check (data_overlap.py); not used for generation.
ann-pt-summ v1.0.1	Credentialed JSONL/XML	Main-study corpus (110 MIMIC-IV hospital courses paired with the	2.3 MB used (archive incomplete: 934 MB of >3.4 GB; the 14	DUA accepted; downloaded 5/12/2026. Used for the judge

Source	Type	Content	Size	Access status and location
		discharge instructions their clinicians wrote) and expert span annotations of unsupported facts: 100 doctor-written and 100 LLM-generated patient summaries plus 10 validation summaries; 423 agreed spans in 10 label types (PhysioNet doi 10.13026/gedc-j464)	annotation and derived files verified against SHA-256)	validation and selection and as the main-study corpus, with local models only. Generations, claims and labels in results_private/ (git-ignored).
MedNLI v1.0.0	Credentialed JSONL	14,049 clinician-written premise and hypothesis pairs from MIMIC-III notes (11,232 train, 1,395 dev, 1,422 test; entailment, neutral, contradiction) used to fine-tune the judge (PhysioNet doi 10.13026/C2RS98)	9 MB	Data use agreement accepted and downloaded 9/15/2026; stored locally (~/Desktop/Thesis/mednli); never committed

Compliance. CITI Program courses 'Data or Specimens Only Research' and 'Conflicts of Interest' completed April 10, 2026 (records 76432215 and 76432216). PhysioNet credentialed data and any output containing text derived from them are never committed, never uploaded to shared cloud storage, and never transmitted to the OpenAI API. Only MTSamples-based generations were sent to the API.

5. Result files (results/)

File	Content
claims_all.csv	2,220 rows: one per claim and condition; columns doc_id, condition, specialty, description, claim, evidence (top-3 sentences, '|'-separated), evidence_scores, label, p_entailment, p_contradiction, is_header. 162 rows are header lines (is_header = True) and are excluded from all metrics; 2,058 claims remain ({'E0': 522, 'E1': 447, 'E1b': 510, 'E2': 242, 'E3': 337}).
summaries.csv	50 rows: source note metadata, the E0/E1/E2 summary texts and word counts, and per-condition UFR, CR and claim counts.
comparison_per_sample.csv	50 rows: per-document UFR, CR and claim counts for every condition, plus paired deltas.
aggregate_statistics.csv	Mean, SD, median per condition; E1−E0 and E2−E0 deltas, % improved, Wilcoxon p, bootstrap 95% CI.
pairwise_tests.csv	Every condition pair × metric: means, medians, mean and median delta, bootstrap CI, W, p, d_z, rank-biserial, % improved/unchanged/worse.
header_filter_effect.csv, header_filter_tests.csv	Metrics and E1-vs-E0 tests before and after removing header lines.
ablation_thresholds.csv, ablation_topk.csv, ablation_cleaned_evidence.csv, claims_cleaned_evidence.csv	Robustness analyses (Chapter 5).
coverage_per_sample.csv, coverage_summary.csv	Coverage proxy per document and per condition.
e2_error_analysis.csv, error_taxonomy_counts.csv, error_taxonomy_examples.csv	Negation analysis and error taxonomy.
calibration_overall.csv, calibration_by_label.csv, calibration_label_counts.csv, calibration_threshold_sweep.csv, calibration_summary_level.csv, calibration_by_system_means.csv, calibration_variants.csv	Judge validation against expert annotations (aggregate statistics only; no MIMIC-derived text).
examples_fixed_by_rag.csv, examples_persistent.csv	Qualitative example pairs.
judge_candidates.csv	Judge selection study: AUROC, threshold, flag rate, precision,

File	Content
	recall, specificity, F1 and kappa for every candidate judge, evidence mode and sentence subset (Chapter 6).
mimic_summary.csv, mimic_pairwise_tests.csv, mimic_reference.csv	Main study aggregates: per-condition means and medians of UFR, CR, coverage of the clinician's instructions, citation accuracy, words, claims and abstentions; paired tests against E0, E1 and the doctor-written instructions; the judge's rates on the doctor-written instructions next to the expert span counts (Chapter 7). No MIMIC-derived text.
mimic_judge_agreement.csv, mimic_second_judge_tests.csv	Agreement between the selected judge and Bespoke-MiniCheck-7B on the 6,588 base-condition claims, per condition, and paired Wilcoxon tests of per-document UFR under the second judge and under both judges.
mimic_ablations.csv, mimic_qa_summary.csv, mimic_qa_utility.csv	Retrieval ablations (paired differences of six E1 variants from the base configuration) and the question-answering pilot aggregates by condition and question type, including the shares of all questions abstained on, answered fully supported and answered with an unsupported sentence.
mimic_condition_ci.csv, mimic_variant_tests.csv, mimic_replicates.csv, mimic_abstention.csv, mimic_equivalence.csv, mimic_window_check.csv, mimic_splitting_check.csv, mimic_tau05/	Revision analyses: per-condition bootstrap intervals, length-matched and no-citation variants, run-to-run variation, abstention, coverage equivalence check, overlapping-window check, splitter comparison, and the main-study aggregates at threshold 0.5.
atomic_results.csv, judge_selection_bias.csv, judge_selection_frequency.csv, judge_per_source.csv, data_overlap.csv	Atomic-fact analysis, optimism and transfer of the judge selection, per-source agreement, and patient and admission overlap (counts only).
audit_summary.csv, audit_types.csv, audit_citations.csv, audit_agreement.csv, audit_*_v1.csv, protected_text_scan.csv, protected_text_scan_summary.csv	Manual audit aggregates for the second labeling pass, the first pass's aggregates under _v1 names, agreement between the passes, and the result of the protected-text scan.
mimic_summary_llama31.csv, mimic_pairwise_tests_llama31.csv, mimic_reference_llama31.csv, and the same for mistral7b	Generator-robustness aggregates: the main-study measures and paired tests recomputed on summaries generated by Llama-3.1-8B-Instruct and Mistral-7B-Instruct-v0.3 (Section 7.10 of the thesis). No MIMIC-derived text.
fig_*.png	19 figures at 200 dpi used in the thesis.

6. Pipeline architecture

- Stage 1, generation: E0 prompts GPT-4o-mini with the first 4,500 characters of the note; E1 prompts it with the top-3 retrieved five-sentence chunks only (query = description + first 300 characters; all-MiniLM-L6-v2 cosine; at most 4,000 characters); E1b adds the full note to the same excerpts; E3 verifies each E1 claim against the top-3 sentences retrieved from the full note and rewrites the summary; E2 selects the top-5 centroid sentences without an LLM.
- Stage 2, claim segmentation: spaCy en_core_web_sm sentences of at least 10 characters; lines that are only a markdown/section header are removed (preprocessing.is_markdown_header).
- Stage 3, evidence retrieval: for each claim the 3 most similar source sentences by cosine similarity of all-MiniLM-L6-v2 embeddings.
- Stage 4, NLI classification: cross-encoder/nli-MiniLM2-L6-H768 scores each (evidence sentence, claim) pair; the claim's p_entailment and p_contradiction are the maxima over the three pairs. Supported if p_entailment ≥ 0.5 and p_entailment > p_contradiction; Contradicted if p_contradiction ≥ 0.5 and p_contradiction > p_entailment; otherwise Not-Supported.
- Stage 5, aggregation: per summary, UFR = (Contradicted + Not-Supported) / claims and CR = Contradicted / claims.
- Stage 6, statistics: two-sided Wilcoxon signed-rank tests on paired per-document metrics, percentile bootstrap 95% CIs (10,000 resamples), d_z and rank-biserial effect sizes, Holm adjustment over all twenty pilot pairwise tests (thesis/results_loader.py); in the main study, over the twelve primary tests of E1, E1b, E2 and E3 against E0 on UFR, CR and coverage (mimic_evaluate.py) and within each measure for the ablations.

- Validation: the unchanged judge is applied to 210 expert-annotated summaries; sentence-level agreement with expert spans, threshold sweep, AUROC, per-label recall, summary-level Spearman correlation, and five evidence-aggregation variants (calibration_annptsumm.py, calibration_variants.py).
- Judge selection: six candidate judges in two evidence modes are scored on the same 1,781 sentences with thresholds chosen on the other subset; the strongest general model is fine-tuned on MedNLI and scored again; the candidate with the highest kappa becomes the main-study judge (judge_candidates.py, finetune_mednli.py).
- Main study, generation: Qwen2.5-7B-Instruct (4-bit, mlx-lm, localhost) writes four-part patient summaries of 110 MIMIC-IV hospital courses under E0, E1 (five-sentence chunks, top-3 dense retrieval, citations required), E1b, E2 and E3, plus six E1 variants (3- or 8-sentence chunks, top-2 or top-5, BM25, no citations); the retrieval query is a fixed section list plus the first 300 characters (mimic_generate.py).
- Main study, evaluation: the selected judge scores every claim against the whole course in windows (UFR, CR), cited excerpts (citation accuracy) and, in the reverse direction, every sentence of the clinician's discharge instructions against the summary (coverage); the doctor-written instructions are scored as a human reference; paired Wilcoxon tests against E0, E1 and the reference, and against the E1 base configuration for the ablations (mimic_evaluate.py).
- Question-answering pilot: three questions per course (medications, follow-up, warning signs) answered from the full course or from three retrieved chunks with an abstention instruction; abstention rate, UFR/CR of answered questions, and support of the answers by the clinician's instructions (mimic_qa.py).

7. How to reproduce

```
git clone https://github.com/slay-a/medical-hallucination-eval.git && cd medical-hallucination-eval
bash setup.sh                       # creates .venv, installs pinned dependencies, downloads en_core_web_sm
# place mtsamples.csv one directory above the repository
export OPENAI_API_KEY='sk-...'      # needed only for generation (E0, E1; optionally E1b, E3)
bash run.sh                         # generation + evaluation + recomputation + ablations + figures
bash run.sh --offline               # reproduce every pilot number from the stored results without any API call
bash run.sh --main-study            # judge study, MedNLI fine-tuning, MIMIC-IV main study and QA pilot (credentialed data, local models)
python thesis/build_thesis.py       # rebuild the thesis .docx and .pdf
```

Changes since the May 2026 description: header lines are excluded from the claim set (162 of 1,373 stored E0/E1/E2 claim rows), which changes every metric; the E1 description was corrected (excerpts only, non-overlapping five-sentence chunks); two new conditions, E1b and E3, were generated and evaluated; four dependencies that the analysis scripts import (matplotlib, scipy, python-docx, reportlab) were added to requirements.txt; bootstrap confidence intervals, effect sizes and Holm adjustment are now actually computed; the ann-pt-summ DUA is signed and the annotation files are in use. Since the September 4, 2026 description: a judge selection study replaced the pilot judge with a DeBERTa-v3-large NLI model validated against the experts, a MedNLI fine-tuning script was added, and the main study on 110 MIMIC-IV hospital courses with a locally served Qwen2.5-7B-Instruct model (five conditions with citations, coverage against the clinician's instructions, citation accuracy, six retrieval ablations, question-answering pilot) was implemented and run. Since the October 12, 2026 description: citation markers are removed before sentence segmentation and assigned by position, and section headers merged into a sentence are stripped (citations.py), which corrects the citation measures; length-matched and no-citation variants, repeated generations, atomic-fact scoring, a selection-bias analysis, a patient-overlap check, an overlapping-window check, per-condition intervals and a blind manual-audit tool were added; dependencies are pinned with lock files; and a scan confirms that no tracked file contains protected text. Since the November 1 description: the manual audit was relabeled in a second blind pass under a clarified rubric on an extended sample (400 items), with the agreement between the passes computed by the new audit_agreement.py; the four main-study conditions were re-generated and re-evaluated with Llama-3.1-8B-Instruct and Mistral-7B-Instruct-v0.3 through a

MIMIC_TAG mechanism in mimic_generate.py and mimic_evaluate.py; and verifact_eval.py was added for an external check of the selected judge on the VeriFact-BHC clinician labels.